Tectonophysics, 146 (1988) 91–103
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

Paleomagnetism of the Umbrian–Marches orogenic belt

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(Received April 15, 1987; accepted April 28, 1987)

Abstract

Hirt, A.M. and Lowrie, W., 1988. Paleomagnetism of the Umbrian–Marches orogenic belt. In: F.-C. Wezel (Editor), The Origin and Evolution of Arcs. *Tectonophysics*, 146: 91–103.

The Lower Cretaceous Maiolica formation was investigated in a regional paleomagnetic study in order to clarify the role of the Umbrian–Marches region in Western Mediterranean tectonics. This area has been considered as a semi-autonomous microplate by some authors and as an allochthonous overthrust block by others. Oroclinal bending has been inferred from earlier paleomagnetic results of the Scaglia formation. In spite of weak natural remanent magnetizations (NRM), stable characteristic remanent magnetization (ChRM) directions were defined for 31 sites. The site-mean directions are well-grouped ($D = 306°$, $I = 37°$, $\alpha_{95} = 5°$) but are not uniform. Fisher statistics are therefore not applicable. The nonuniformity of directions is partially due to regional and local variations, but is not related to the orientation of the fold axes as suggested by several authors. It is possible that the Umbrian–Marches region has suffered thrust décollement on the underlying Triassic evaporites during the Late Tertiary orogeny, but this was not accompanied by large-scale translation or rotation relative to the assumed autochthonous parts of the Adriatic promontory.

Introduction

The tectonic history of the Northern Apennine mountain belt derived from paleomagnetic studies has been discussed by several authors over the past 10 years (Lowrie and Alvarez, 1974, 1975; Channell and Tarling, 1975; Van den Berg and Wonders, 1976; Van den Berg et al., 1978; Channell et al., 1978; Van der Voo and Channell, 1980; Eldredge et al., 1985). Although these studies addressed the problem of autochthony of the Umbrian–Marches region and the relevance of paleomagnetic directions to the Adriatic promontory, no definitive answer was found.

The earliest paleomagnetic studies in the Umbrian–Marches region concentrated on the Lower Cretaceous Maiolica formation and the Upper Cretaceous to Lower Tertiary Scaglia formation (Lowrie and Alvarez, 1974, 1975; Channell and Tarling, 1975; Van den Berg and Wonders, 1976; Van den Berg et al., 1978). These studies concluded that the Umbrian–Marches region had undergone anti-clockwise rotation with respect to Africa. Lowrie and Alvarez (1974, 1975), Van den Berg and Wonders (1976) and Van den Berg et al. (1978) considered the region to be autochthonous or para-autochthonous, and that Umbria rotated as part of a separate microplate. Channell and Tarling (1975) considered the region to be allochthonous and attributed the anti-clockwise rotation to decoupling on the Triassic evaporates.

A later regional study of the Scaglia formation by Channell et al. (1978) continued to maintain the allochthony of the Umbrian–Marches sequence. The authors suggested that the Umbrian Apennines underwent oroclinal bending, indicated by the curvature of the mountain belt and its fold axes from north to south and the change in paleomagnetic declination from north to south. The northern part of the Umbrian–Marches re-

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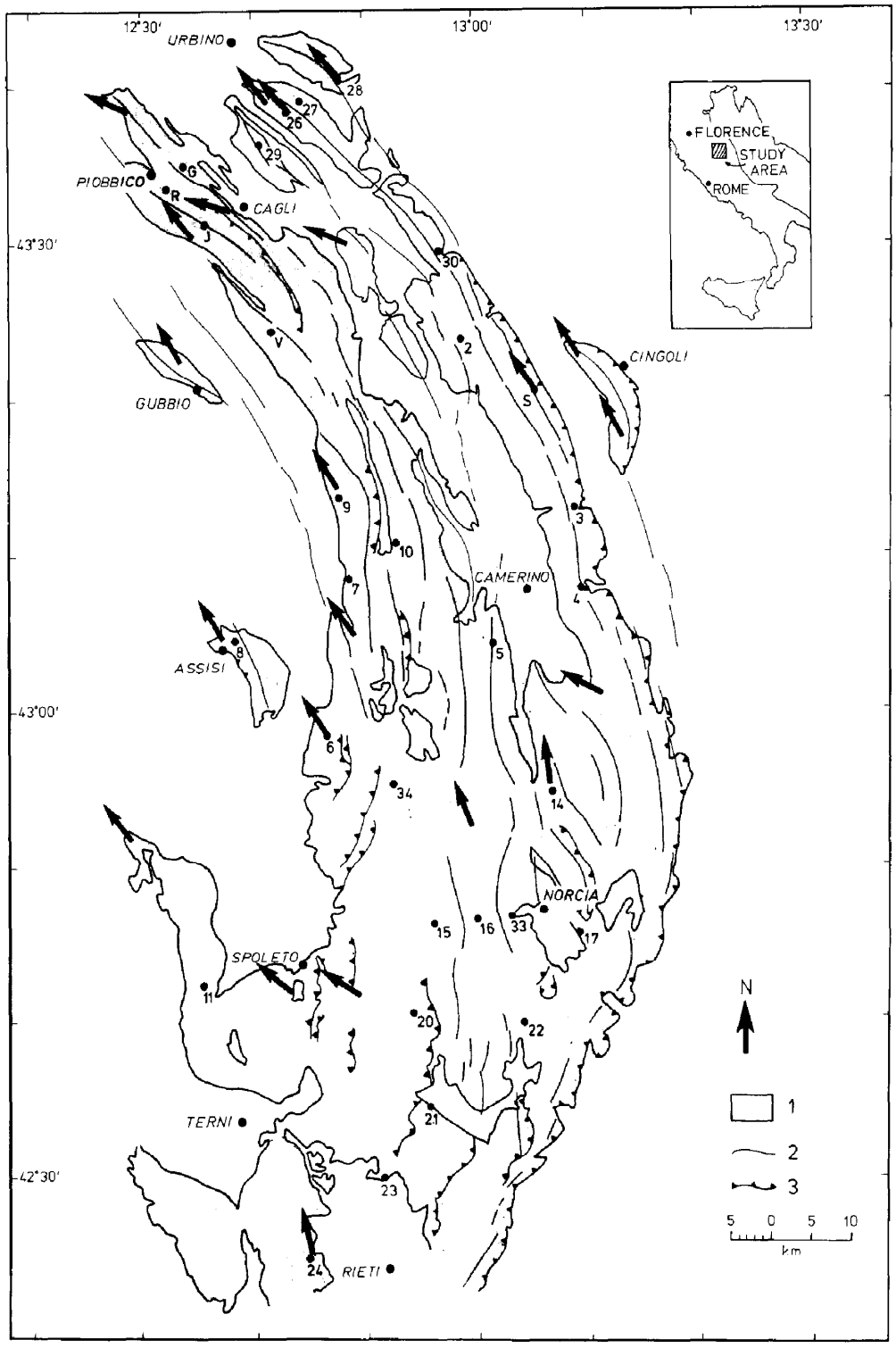


Fig. 1. A schematic tectonic map of the study area. *1* = Lower Jurassic to Middle Cretaceous Calcareous Supergroup; *2* = anticlinal fold axes; *3* = thrust faults. Arrows show paleodeclinations of tectonically corrected site-means of the Scaglia Rossa formation and dots indicate Maiolica formation sites.

gion consists of a series of anticlines striking 315^0 and the mean paleomagnetic declination for the Scaglia formation is 316^0. The southern region has fold axes which trend 350^0 and the mean paleomagnetic declination is 338^0 (Channell et al., 1978). Magnetostratigraphic studies of the Scaglia formation in Gubbio by Lowrie and Alvarez (1977) showed an increasing trend towards more northerly declinations from the Aptian to the Eocene. Each site of the Channell et al. (1978) study was dated paleontologically to show that their results were not dependent on a bias of stratigraphic sampling level.

In a review of paleomagnetism in orogenic belts, Van der Voo and Channell (1980) reexamined the data from Channell et al. (1978). They suggested that the oroclinal bending was not the mechanism causing the dependence of paleomagnetic directions to the strike of the fold axes in the Umbrian arc. The authors proposed that the arcuate fold belt formed by differential horizontal transport accompanied by en échelon faulting, which changes gradually on a décollement surface along the fold belt.

Eldredge et al. (1985) made a more rigorous examination of the Channell et al. (1978) data-set to check for oroclinal bending. This study used a method described in Schwartz and Van der Voo (1983) to determine the relation between fold axes and declinations. It concluded that oroclinal bending had taken place in the Umbrian orogen; the trend of the originally straight belt was closer to the present trend of the northern part of the belt, relative to which the southern part subsequently underwent a clockwise rotation.

However, Eldredge et al. (1985) determined the fold axes strikes and the paleomagnetic declinations in the Umbrian–Marches region from a sketch-map in the Channell et al. (1978) publication. These directions cannot be considered sufficiently accurate for a tectonic study due to their original definition. First, it should be noted that the fold axes orientations in this map are largely schematic. A recent tectonic map of the central Apennines (C.N.R., 1982) shows that the trends of the fold axes are more complicated than in the schematic map. Eldredge et al. (1985) also selected a single reference declination for the Scaglia formation. Since magnetostratigraphic studies show that the paleomagnetic declinations change progressively from Aptian through to the Eocene (Channell et al., 1978), a single reference declination cannot be used for sites the ages of which cover this entire time span (Lowrie and Hirt, 1986).

The demagnetization data of Channell et al. (1978) were generally evaluated using methods employed frequently in the early 1970's to define paleomagnetic directions. The ChRM direction at a site was defined by a minimum dispersion criterion at an optimum demagnetization step. This method does not describe a characteristic vector or a stable end-point, and often secondary components of magnetization have not been completely removed.

The Scaglia data set of Lowrie and Alvarez (1975) was enlarged with six additional sites; several sites of Channell and Tarling (1975) included in Channell et al. (1978) were discarded if they were geologically close but gave very different directions. Only sites which had mean directions with $\alpha_{95} \leqslant 15^0$ were considered suitable for tectonic evaluation (Fig. 1). The number of sites for which sufficiently detailed demagnetization data were available, and which had acceptable α_{95} values, was strongly reduced. The distribution of Scaglia Rossa site-mean declinations no longer shows a distinct dependence on fold axes trend.

The present study was undertaken as a re-evaluation of the paleomagnetism of the Umbrian–Marches region, using more stringent laboratory techniques. The investigation was done on the Lower Cretaceous Maiolica formation. Unlike the Scaglia formation, the Maiolica formation does not show any trends in paleomagnetic declination or inclination through the Early Cretaceous (Lowrie et al., 1980a; Lowrie and Channell, 1983; Cirelli et al., 1984; Lowrie and Alvarez, 1984). Therefore, the age of the Maiolica at a site is not important in comparing declinations within the region.

Geology of the Umbrian sequence

The Mesozoic sedimentary sequence of the Umbrian–Marches region was deposited on the

southern margin of the Tethys ocean. The basal unit of the Northern Apennine mountain chain is the Late Triassic Burano formation, which consists of anhydrites, marls, dolomites and dolomitic limestones. Although the Burano formation is found outcropping only in Tuscany, it is known in the Umbrian–Marches region from the Perugia-2 drill hole (Ghelardoni, 1962). A period of uniform subsidence in the Jurassic led to fault-bounded basins separated by topographic highs, such that local differences in facies and thickness occurred (D'Argenio and Alvarez, 1980). There are three types of stratigraphic sequences found in the region: complete-sequences, characteristic of basins; condensed-sequences, characteristic of fault-bounded topographic highs; and composite-sequences (Colacicchi et al., 1970; Centamore et al., 1971; Koopman, 1983). The basinal sequences remain uniform in their stratigraphy throughout the Northern Apennines. The topographic differences become less extreme by the Late Jurassic with the start of deposition of the Maiolica formation. The Upper Maiolica condensed and complete sequences become laterally continuous stratigraphically (Micarelli et al., 1977), indicating the inactivity of structural boundary faults in the Early Cretaceous. Intercalated limestone turbidites and debris sheets are often found in the eastern and southeastern part of the Umbrian–Marches region (Cantelli et al., 1978; Castellarin et al., 1978; Alvarez et al., 1985).

The Maiolica limestone is a white- to grey-colored, well-bedded limestone with frequent interbedded grey- to black-colored chert beds or nodules. Its lower contact is gradational with the Tithonian Calcare diasprini limestone, and its upper contact with the Late Aptian Fucoid marls. The Maiolica reaches thicknesses of 400 to 500 m in the basinal sequences, but can be only 100 m thick in condensed sequences (Micarelli et al., 1977). Slumping is found in the basinal sequences.

The general structure of the Northern Apennine region is that of an arcuate mountain chain convex to the east-northeast (Fig. 1). The northwestern part of the chain consists of large, box-like fold structures trending NW–SE and facing northeast. Due to their box-like shape, it is difficult to define a single fold axis (Lavecchia, 1981). Folds plunge suddenly at either end at angles of approximately 10°. The southern part of the fold belt consists of N–S anticlines which become spatially closer than in the north. These large anticlines are complex in their structure with reverse and thrust faults cutting their limbs (Giannini and Lazzarotta, 1975; C.N.R., 1982). Some folds in the southern region still trend NW–SE.

Tectonic deformation developed in the Late Oligocene through the Miocene and at least into the Pliocene (Reutter and Groscurth, 1978). The characteristic fold geometry suggests that the deformation is limited to the sedimentary cover (Giannini and Lazzarotta, 1975; Coli, 1981; Lavecchia, 1981, 1986; Koopman, 1983). The large scale thrust motion during the Tertiary often used pre-existing Jurassic normal fault planes as ramps, which trend NW–SE or NNE–SSW (Koopman, 1983).

Methodology

Approximately 800 specimens were collected at 34 sites for this paleomagnetic study throughout the Umbrian–Marches region (Fig. 1). Sites were not taken where the fold axes die out and the plunge can be appreciable; therefore, corrections for plunging fold axes were not necessary. A minimum of sixteen samples were drilled at each site using a portable gasoline drill. Five sites were block-sampled and later drilled out in the magnetic laboratory.

The NRM was measured on a three-axis cryogenic magnetometer interfaced with a HP-1000 minicomputer. Three or six replicate measurements were made for each specimen, overdefining the remanence vector such that repeatability of each measurement was used as a rejection criterion (Lowrie et al., 1980b). A lower sensitivity limit of 3×10^{-6} A m^{-1} is achieved for the cryogenic system.

Thermal demagnetization was done using a Schonstedt thermal demagnetizer with a rest field less than 5 nT, and alternating field (AF) demagnetization using a single-axis Schonstedt geophysical demagnetizer. The magnetic susceptibility was monitored after each thermal demagnetization step with a KLY-1 bulk susceptibility bridge. The

TABLE 1

Paleomagnetic site-means of the Maiolica formation *

Site	$s(°)$	n	$D(°)$	$I(°)$	k	$\alpha_{95}(°)$
MJ2	338	14	290.2	37.8	47.4	5.8
MJ3	344	15	314.0	35.0	35.7	6.5
MJ4	342	11	325.7	32.2	48.3	6.6
MJ5	359	13	308.1	50.3	47.9	6.1
MJ6	316	14	142.3	−32.7	63.8	5.0
MJ7	358	15	286.8	40.5	84.1	4.2
MJ8	335	14	129.4	−36.5	51.7	5.6
MJ9	349	14	131.8	−32.5	37.9	6.5
MJ10	350	16	298.3	35.1	51.4	5.2
MJ11	–	15	146.4	−32.0	59.4	5.0
MJ14	340	6	337.4	36.3	45.2	10.1
MJ15	012	16	314.0	34.4	39.6	5.9
MJ16	337	10	144.4	−30.3	67.4	5.9
MJ17	–	14	123.9	−27.5	31.6	7.2
MJ20	320	16	331.1	40.9	50.3	5.3
MJ22	–	13	309.2	42.9	58.9	5.4
MJ23	–	7	280.2	41.2	49.2	8.7
MJ24	–	13	307.9	34.2	61.5	5.3
MJ26	308	8	309.2	35.5	18.6	13.2
MJ27	300	10	124.2	−37.0	32.1	8.7
MJ28	315	9	101.8	−24.3	40.0	8.2
MJ29	315	11	306.0	40.2	56.7	6.1
MJ30	326	13	114.9	−35.8	55.3	5.6
MJ32	321	10	148.1	−34.9	40.4	7.3
MJ33	337	5	298.4	25.0	56.1	10.3
MJ34	–	10	340.4	29.6	62.7	6.2
R	304	21	284.4	42.0	35.6	5.4
S	335	27	297.3	33.5	104.3	2.7
V	314	10	289.1	33.7	57.6	6.4
G	313	25	269.3	41.1	56.3	3.9
J	305	18	284.3	32.7	59.9	4.5

* Columns give the site number, strike of the fold axis (s) where it could be defined, number of specimens (n), declination (D) and inclination (I) of site-means after tectonic tilt correction, k and circle of 95% confidence (α_{95}), R, S V, G and J are data from magnetostratigraphic sections

acquisition of isothermal remanent magnetization (IRM) was observed in fields up to 3.5 T, using a cryogenic magnet; the IRM was then demagnetized thermally. The coercivities demonstrated by the acquisition curves were used in conjunction with the thermal demagnetization behavior to interpret magnetic carriers.

The ChRM direction for specimens was resolved from orthogonal progressive demagnetization diagrams (Zijderveld, 1967; Dunlop, 1979). Vectors were calculated with principal component analysis and a two dimensional least squares fit method. The ChRM vectors defined by both methods were invariably equivalent. ChRM directions for five earlier magnetostratigraphic sections (Lowrie et al., 1980a; Lowrie and Channell, 1983; Lowrie and Alvarez, 1984) were re-evaluated using the least-squares method (Table 1).

Results

NRM intensities for the Maiolica formation were weak, ranging from 3×10^{-3} to 3×10^{-6} A m^{-1}, but stable and repeatable. Detailed pilot investigations of thermal demagnetization behavior were performed on at least two specimens from each site. The majority of the specimens from 29 sites were demagnetized thermally, which

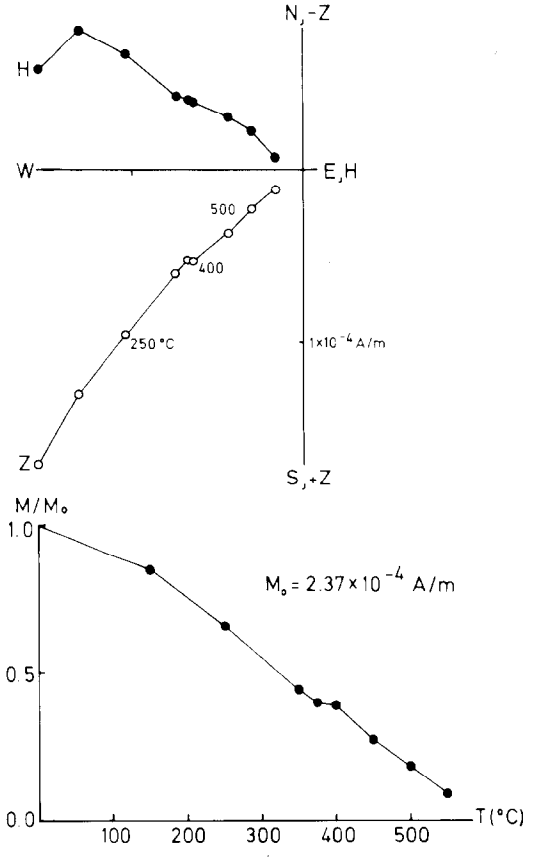


Fig. 2. Representative vector diagram which shows directional variation and normalized intensity variation during thermal demagnetization after tilt correction. Closed (open) circles are used for the horizontal (vertical) projection for this and subsequent figures.

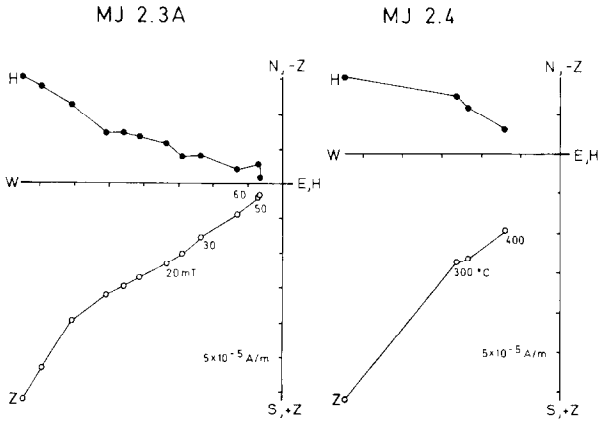


Fig. 3. Vector diagrams illustrating similar vector components defined in alternating field and thermal demagnetization after tilt correction.

revealed a distributed blocking temperature spectrum. A low temperature component of spurious direction was removed by 300 ° C. The characteristic remanent magnetization (ChRM) was defined between 300 to 550 ° C (Fig. 2).

AF demagnetization was also applied to at least two specimens from each site. Median destructive fields (MDF) were between 5 to 20 mT for the majority of the specimens, although some MDF's reached 50 mT. Ninety-five percent of the intensity was removed from 20 to 80 mT. Both AF and thermal methods isolated similar ChRM vectors for a site (Fig. 3). Demagnetization vectors decay close to the origin, but not always directly to the origin. This indicates a hard coercivity or high temperature component still remains. Since this component accounts for less than 5% of the NRM intensity, it is generally within the instrumental noise level of the cryogenic magnetometer and cannot be properly evaluated.

IRM acquisition reveals both a low coercivity and high coercivity component (Fig. 4). The low coercivity component, shown by a kink in the IRM curve at 0.12 T, is probably due to magnetite. The presence of magnetite is also indicated in thermal and AF demagnetization. The higher coercivity component could result from hematite or goethite. Hematite is the more likely magnetic carrier since the curve appears to be starting to saturate by 3.5 T. If goethite were present, the IRM would still increase in intensity at these fields (Lowrie and Heller, 1982).

Specimens given a 3.5 T IRM along their vertical axis later had a 0.1 T field applied along their *X*-axis. Thermal demagnetization of the saturation IRM shows that the 0.1 T component does not disappear until the specimen has been heated to 580 ° C (Fig. 5a, b). This lower coercivity compo-

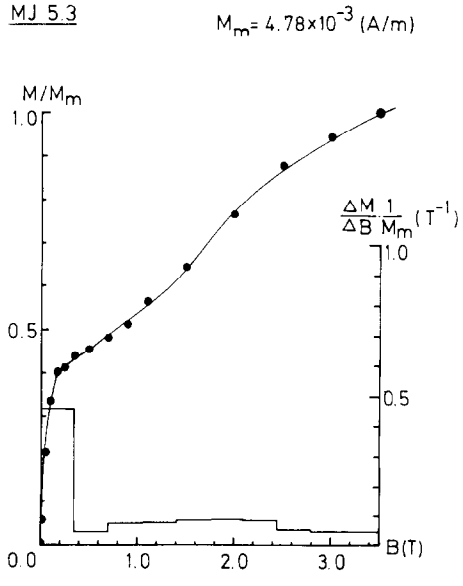


Fig. 4. IRM acquisition curve for a representative sample of the Maiolica formation.

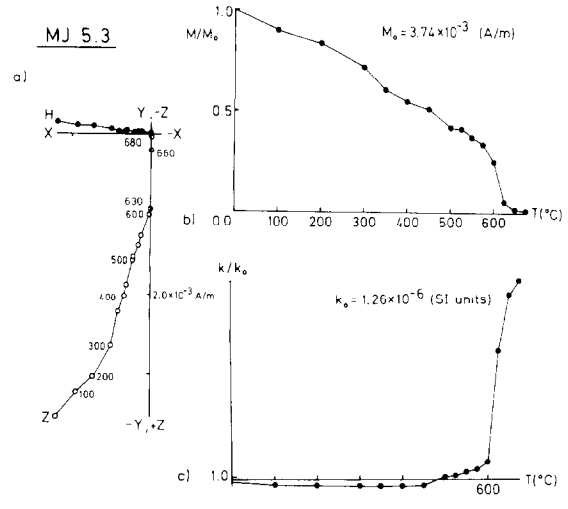


Fig. 5. Thermal demagnetization of the saturation IRM: (a) Vector diagram; (b) normalized intensity plot; and (c) change of bulk susceptibility with heating.

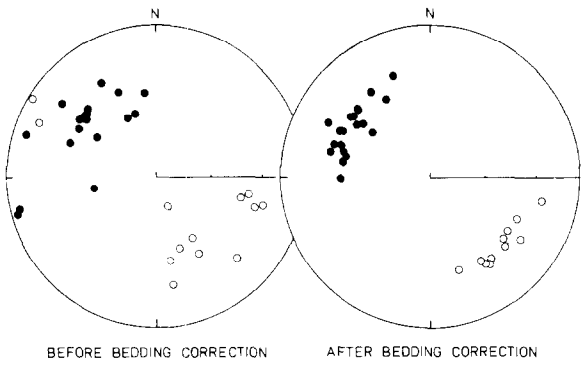


Fig. 6. Equal area plot of ChRM directions for individual site means before and after tectonic tilt correction. Closed symbols are plotted on the lower hemisphere and open symbols on the upper hemisphere for this and subsequent figures.

nent is probably carried by a magnetite fraction. The 3.5 T component is probably carried by hematite, since it exists until 680 °C.

Monitoring the bulk susceptibility with heating shows that the specimens are initially diamagnetic. Between 450 to 500 °C the susceptibility becomes positive, which indicates that ferromagnetic material is being created in the heating process (Fig. 5c).

The ChRM directions defined for each site are shown in Fig. 6 and are listed in Table 1. Eight sites were discarded because there was large within-site scatter of their directions ($a_{95} > 15^0$). The data set includes newly defined directions for the six magnetostratigraphic studies. A test of the applicability of Fisher statistics shows that the inclinations conform to a Fisherian distribution, but that the declinations are not azimuthally uniform. Therefore Fisher statistics should strictly not be applied to the data. For this reason we calculate separately the 95% fiducial ranges of inclination and declination, given as: $\pm t\sigma/\sqrt{N}$, where σ is the appropriate standard deviation, N the number of observations, and t the corresponding value of Student's t-statistic for $(N-1)$ degrees of freedom. The ChRM directions show a statistically significant improvement upon applying a simple tectonic tilt correction. The bedding-corrected mean regional direction for the 31 Maiolica sites is: $D = 306.4^0$, $I = 36.7^0$, $e_D = \pm 6.6°$, $e_I = \pm 2.0°$, where e_D and e_I are the 95% fiducial errors in declination and inclination, respectively.

Discussion

Magnetostratigraphic studies of the Maiolica formation by Lowrie and Alvarez (1984, fig. 4) have shown that there are no significant trends of the declination or inclination at the 95% confidence level within a section. The azimuths of the magnetic directions within the magnetostratigraphic sections as well as in the regional study are smeared. To test whether the declination and inclination dispersion in the magnetostratigraphic section was comparable to that seen in our regional site-means, the standard deviations of the individual declination and inclination distributions were computed.

Curray (1956) has discussed statistics of circular distributions. The preferred orientation of a $0–360^0$ distribution is equated with the vector mean, calculated in the usual manner by summing the components of the individual vectors. The mean of a $0–180^0$ distribution is found by doubling the angles and again finding the mean from the vector sum. The magnitude of the vector sum is a measure of the data dispersion, and gives a good correlation with the standard deviation about the vector direction, which we have used.

Four sections, at Gorgo a Cerbara (G), Bosso (J), Presale (R) and Frontale (S), were divided into subsites of twelve specimens, representing approximately 8 m of section. The standard deviations of the subsite declinations and inclinations are shown in Table 2. The difference between the standard deviations of the declination and inclination is not statistically significant at the 95% confidence level for sections J, R and S, and is only marginally significant for section G.

Similarly, the between-site standard deviations of the declinations and inclinations, respectively, have been computed for the regional distribution of Maiolica paleomagnetic sites. In this case the standard deviations are statistically different at the 95% confidence level. The standard deviation of the inclinations is comparable to that seen in the magnetostratigraphic sections, but the standard deviation of the declinations is about three

TABLE 2

Standard deviations of declination (D) and inclination (I) *

	G	J	R	S	MJ	MJn	MJs
Standard dev. $D(°)$	10.2	5.1	6.1	3.1	18.2	12.2	16.0
Standard dev. $I(°)$	5.7	5.4	5.1	5.3	5.4	4.4	6.3

* Columns give magnetostratigraphic section Gorgo a Cerbara (G), Bosso (J), Presale (R) and Frontale (S), mean direction of the Maiolica formation (MJ), mean direction for sites north of Assisi-Cingoli (MJn) and sites south of Assisi-Cingoli (MJs).

times as large (Table 2). This suggests that the between-site declination scatter in the Maiolica regional study may be due to uncompensated tectonic effects.

Channell et al. (1978), Van der Voo and Channell (1980) and Eldredge et al. (1985) suggested that the ChRM directions of the Scaglia formation were related to the fold axes. Could the nonuniformity of the tectonically corrected ChRM directions of the Maiolica formation also be due to their dependence on local fold axis orientation?

One method to examine the relationship between site-mean directions and the local fold axes is to rotate all the fold axis strikes to correspond to a common strike direction for the entire Umbrian–Marches region. The strikes of fold axes at our sites were distributed fairly uniformly between 300^0 and 360^0. The amount of rotation applied to the fold axis at each site in order to align the strike of the local axis with the reference strike is also applied to the ChRM declination. If the fold belt was bent after primary folding, then the grouping of ChRM directions should improve after rotating the fold axes with their accompanying remanent declinations. Figure 7 shows the distribution of directions after compensating for the deviations of the local fold axes from an arbitrarily chosen regional trend. This test was only made at the 25 sites where a regional fold axis could be defined. The ChRM directions for these sites show a slight increase in dispersion of the declinations after unbending the fold belt ($D = 294^0$, $I = 37^0$, $e_D = \pm 9.0$, $e_I = \pm 2.2$). However, the increase is not statistically significant and cannot be interpreted as evidence that the arcuate shape of the fold belt is primary.

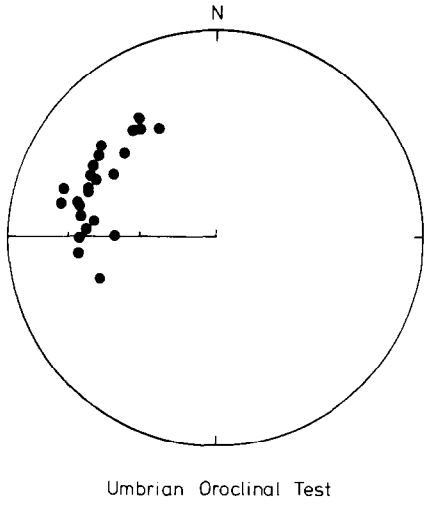


Fig. 7. Equal area plot of the ChRM directions of site-means rotated to account for variation of the local fold axes from an arbitrarily chosen regional trend.

The relative variations of paleomagnetic declinations and fold axis strikes about their respective means provide another method of testing their dependence (Schwartz and Van der Voo, 1983; Eldredge et al., 1985; Lowrie and Hirt, 1986). At any site the deviation of declination from its expected value should correspond to the deviation of the local fold axes from a regional trend. Plotting the declination deviations against the deviations of the local fold axis strike should yield a line of positive unit slope if the mountain belt was bent after the ChRM directions were acquired. Figure 8a shows the declination deviations for each site plotted against the deviations of the local fold axes from the average regional strike. It should be noted that these local fold axes are large-scale structures with an extent of 5 to 10 km. If a representative fold axis could not be defined on this scale, the site was excluded. A reference declination for the Maiolica formation is the re-

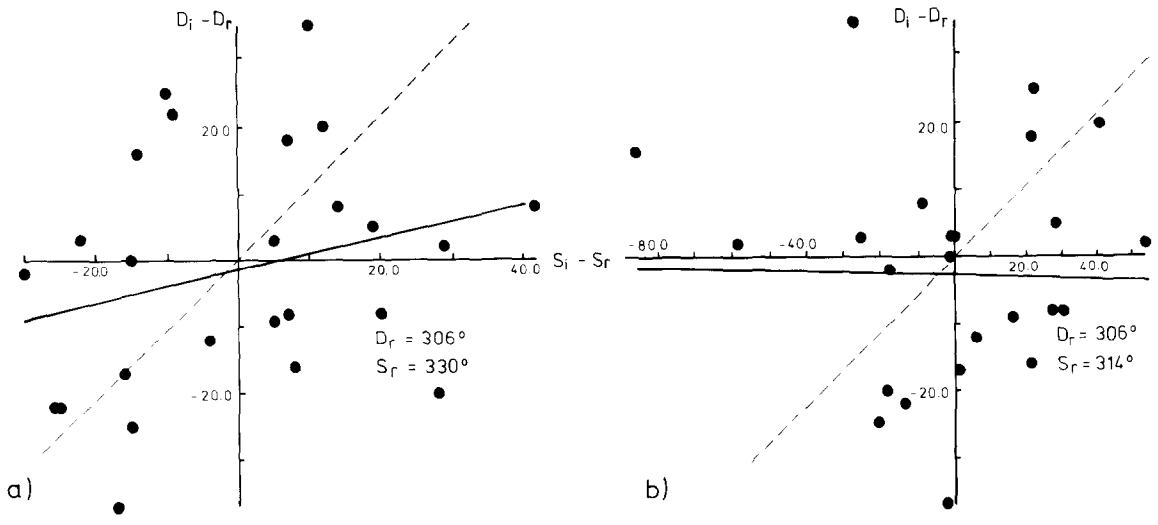


Fig. 8. Diagrams showing declination deviations relative to: (a) fold axis strike deviations and (b) bedding strike deviations for the Umbrian–Marches region. D_r is the reference declination and D_i is the observed declination at a site; S_r is the reference direction of the fold axes (a) or the bedding strikes (b) for the region and S_i is the fold axis direction (a) or bedding strike (b) at a site. The dashed line with unit slope represents oroclinal bending of an originally straight fold belt. The heavy solid line is a best-fit line calculated by linear regression analysis.

gional site-mean for this study, 306^0. The reference strike of the fold axes was 330^0. The slope of the regression line in Fig. 8a is not significantly different from zero slope at the 95% confidence level, as tested with the statistical t-test.

We have also tried correlating the declination variation with that of the bedding strikes (Fig. 8b), assuming that the folds are cylindrical on first approximation. This is equivalent to comparing fold axes on a very local scale. This correlation also leads to a slope which is not statistically different from zero slope at the 95% confidence level.

Although the ChRM directions are not simply related to the fold axes, regional differences can be found within the data. Sites north of a line from Asissi to Cingoli tend to have declinations more westerly than the mean declination of 306^0 (Fig. 9). Sites south of this line have declinations which are more northerly. The separate site means are $D = 295^0$, $I = 37^0$ ($e_D = \pm 6.8$, $e_I = \pm 2.5$, $N = 15$) for the northern group, and $D = 318^0$, $I = 36^0$ ($e_D = \pm 8.3$, $e_I = \pm 3.4$, $N = 16$) for the southern group. The northern site directions are only marginally uniformly distributed about their means and the southern sites are not azimuthally uniform. Again both groups have a scatter of the declinations which exceeds that seen in the magnetostratigraphic sections (Table 2). We have separately plotted the declination variations against the variations in strike of the fold axis for both the northern and the southern groups of data. In both regions the slopes are not significantly different from zero-slope. Although we were only able to define fold axes at nine of the southern sites, the geographic distribution was adequate to be representative of the southern region. The between-site scatter in both groups suggests that the cause of the directional scatter is similar in both the northern and southern sites.

Thrust faults cut through to the surface in the southern part of the fold belt, and the folds are also cut by normal faults. The folding and faulting are very complicated. In the northern part blind thrusting may occur (Koopman, 1983; D. Roeder, pers. commun., 1984), and normal faulting is not as predominant. The scatter of the paleomagnetic declinations may result from these local tectonic variations, which cannot be accounted for at the present time without more detailed structural mapping.

A possible explanation for the curvature of the mountain belt is that it results from ramp-thrust tectonics. Geologic field evidence indicates that

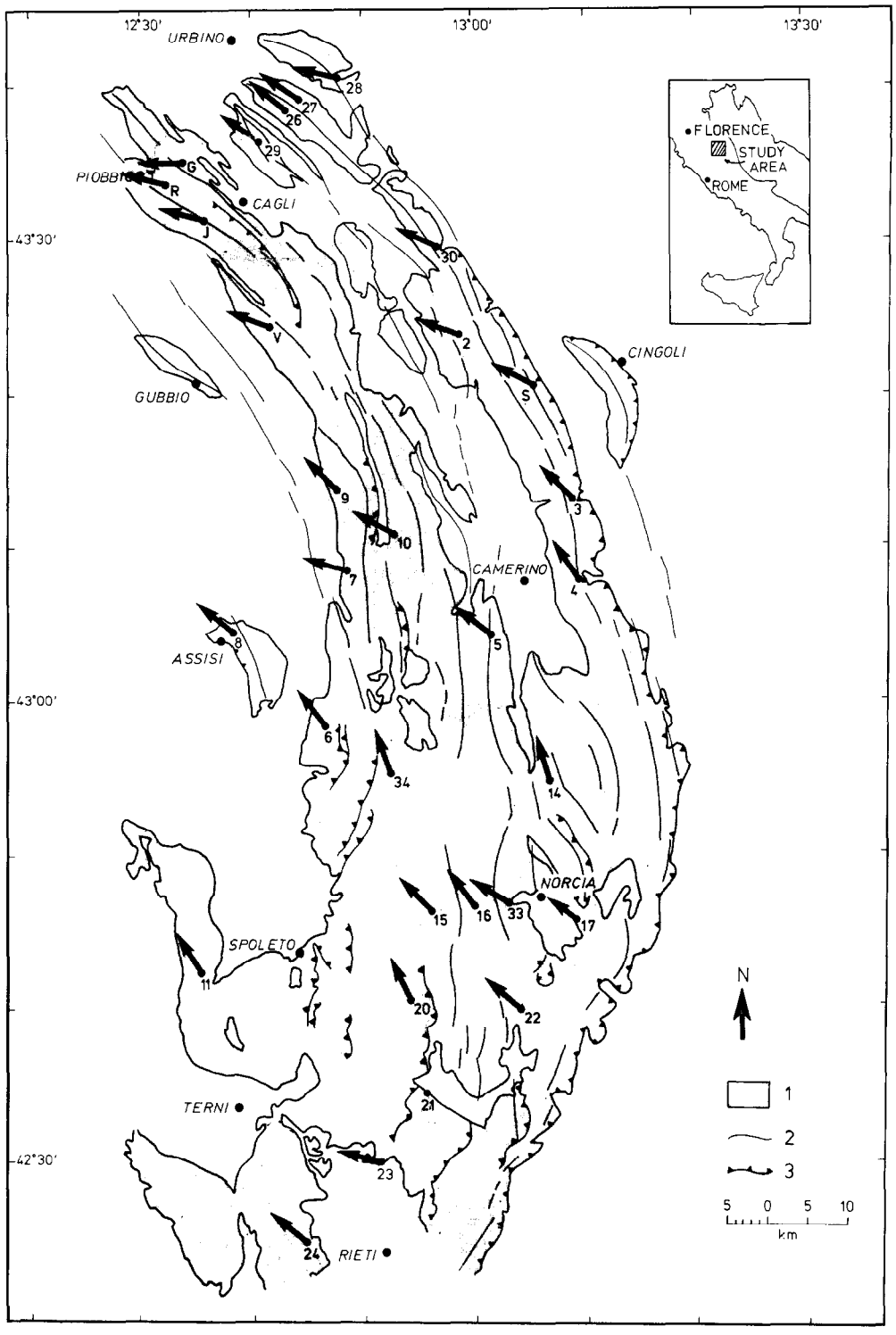


Fig. 9. Paleodeclinations of the Maiolica formation. Symbols are the same as in Fig. 1.

the thrust direction throughout the Umbrian–Marches region is towards the northeast. When this sheet moves over a ramp trending NW–SE the resulting fold has a trend of NW–SE, since the thrust direction is normal to the trend of the ramp. If the trend of the ramp is oblique to the

TABLE 3

Paleomagnetic directions for Lower Cretaceous rocks from the Adriatic promontory *

	N	$D(°)$	$I(°)$	k	$\alpha_{95}(°)$
Umbria region					
This study	31	306.4	36.7	27.0	5.1
Van den Berg et al. (1978)	10	298.6	47.6	–	8.0
Cirilli et al. (1984)	3	280.5	31.3	158.3	9.8
Gargano					
F. Horner (unpublished)	3	309.9	30.5	198.9	8.8
Van den Berg (1983)	8	303.1	35.1	–	8.7
Istria					
Marton and Veljovic (1983)	29	307.2	40.5	–	6.9

* Number of sites (N), remaining columns as in Table 1

direction of overthrust, then there is a dextral strike-slip component of displacement. The resulting fold is no longer related to the direction of overthrusting. The curvature of the fold belt results from the interference between overthrusting in directions perpendicular and oblique to the underlying ramps and their resulting folds (Koopman, 1983).

The role of the Umbrian–Marches region in interpreting the regional tectonic history of the Adriatic has been diversely interpreted (Channell et al., 1979; Lowrie, 1980, 1986; Van den Berg and Zijderveld, 1982; Marton and Marton, 1983). Independent seismic and geologic field evidence from the Umbrian–Marches region indicate that the mountain belt is allochthonous (D. Roeder, pers. commun., 1983; Lavecchia et al., 1984; Lavecchia, 1986). If this is so, paleomagnetic evidence from the Northern Apennines should be excluded from plate reconstructions. However, the Maiolica site-mean direction from the present study agrees well with other Lower Cretaceous data from the Adriatic promontory (Fig. 10, Table 3). Paleomagnetic studies which included data only from Northwestern Umbria do not agree as well with the data from Gargano or the Istrian peninsula, which are presumed to be autochthonous. The directions of Van den Berg et al. (1978) differ from all other studies in the area by having steeper inclinations: however, the data of Cirelli et al. (1984) show a declination discrepancy which reflects the regional variation of declination seen in the Umbrian–Marches fold belt. The limited geographic distribution of sites used in these cited studies implies that the data are not representative of the Northern Apennine region as a whole.

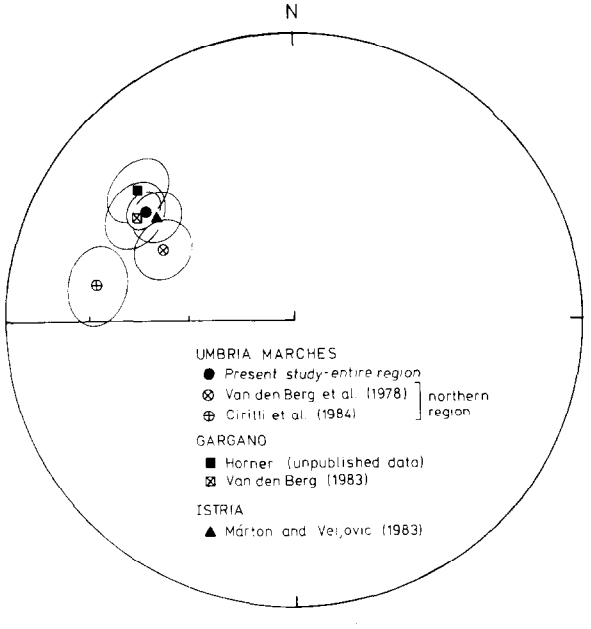


Fig. 10. Equal area plot of regional paleomagnetic site-means of Lower Cretaceous age with circle of 95% confidence for assumed autochthonous areas of the Adriatic promontory.

Conclusions

The mean paleomagnetic direction for all the Maiolica sites is in good agreement with the directions of presumed autochthonous regions of the Adriatic promontory (Lowrie, 1986). This agree-

ment suggests that the Umbrian–Marches allochthon has undergone only a simple translation without major rotation relative to the basement.

Paleomagnetic data from the Umbrian–Marches region do not support the interpretation of oroclinal bending. The curvature of the Northern Apennines is more likely due to the ramp-thrust tectonics characteristic to the region. However, the results do suggest that there is a regional N–S difference in the data, but that the mechanism responsible for the between-site scatter of the declinations is in operation throughout the entire fold belt.

Acknowledgements

We wish to thank A. Koopman, W. Alvarez, G. Pialli and G. Lavecchia for helpful discussions. D.V. Kent and R. Van der Voo are thanked for their critical reviews. This research was funded through the Swiss National Science Foundation, grant 2.684-0.82. Contribution No. 532, Institut für Geophysik, ETH Zürich.

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